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EXP NO: 1 DATE:	AZURE DEVOPS ENVIRONMENT SETUP
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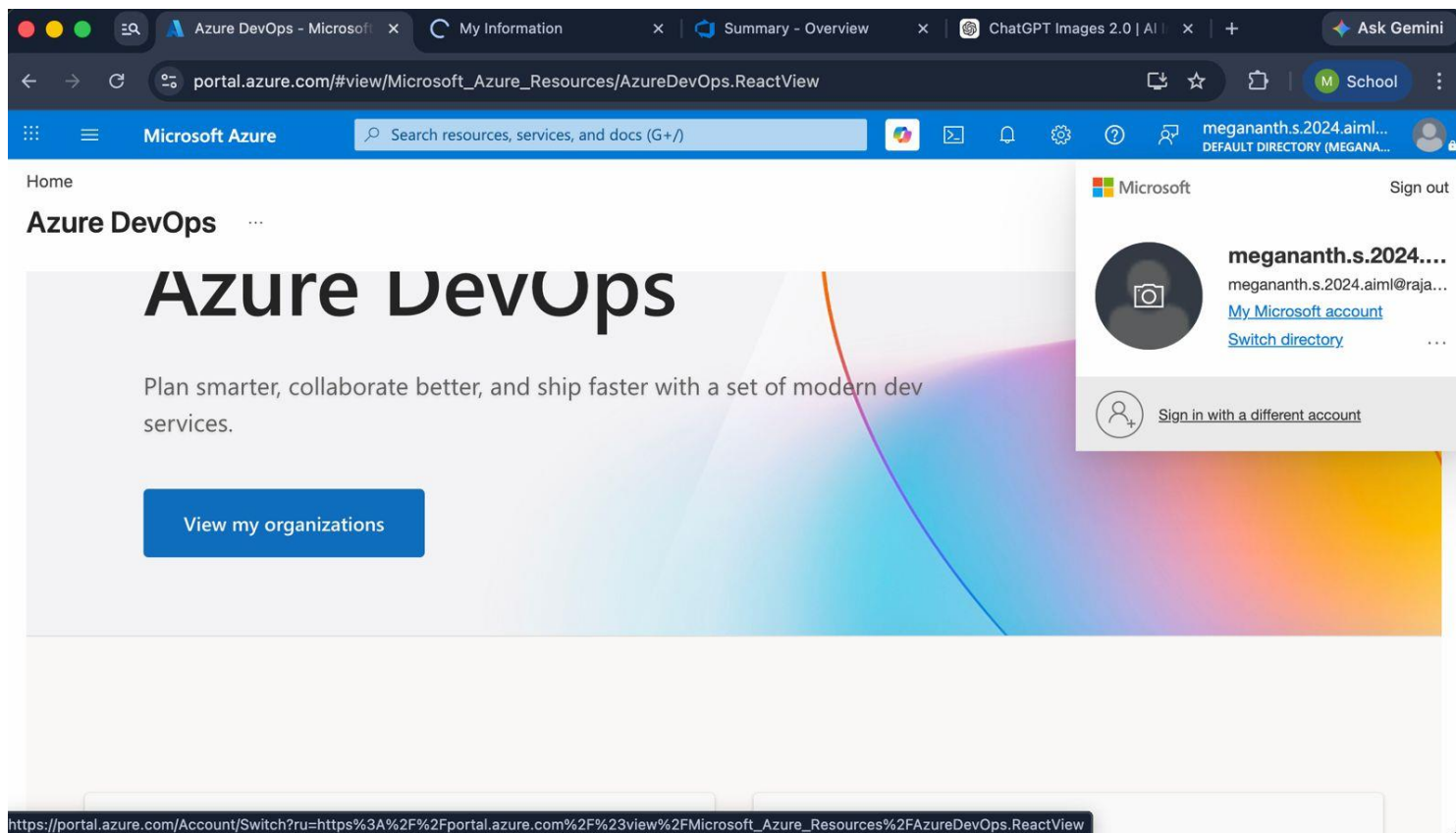
AIM:-

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION:-

1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/get-started/azure-portal>. Sign in using your Microsoft account credentials.

If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



2. Azure home page

The screenshot shows the Azure DevOps web interface in a browser. The browser's address bar displays `aex.dev.azure.com/me?mkt=en-GB`. The user's profile on the left is for Megananth S, with email `megananth.s.2024.aiml@rajalakshmi.edu.in` and a Microsoft account. The main section, titled "Azure DevOps Organizations", lists the user's organizations and projects. The first organization, `dev.azure.com/megananths2024aiml` (Owner), contains three projects: "crime rate prediction", "crime rate", and "crime-rate -prediction". A "New project" link is available. Other organizations listed are `dev.azure.com/abrinrithish` (Member) and `dev.azure.com/mukeshdharman` (Member). A "Create new organization" button is in the top right. At the bottom, there is a "Visual Studio Dev Essentials" banner.

Microsoft

MUKESHDHARMAN | (rajalakshmi.edu.in)

Azure DevOps Organizations [Create new organization](#)

dev.azure.com/megananths2024aiml (Owner)

Projects

- crime rate prediction
- crime rate
- crime-rate -prediction

[New project](#)

Actions

[Open in Visual Studio](#)

dev.azure.com/abrinrithish (Member)

dev.azure.com/mukeshdharman (Member)

Visual Studio Dev Essentials

Get everything you need to build and deploy

- Open DevOps environment in the Azure platform by typing **Azure DevOps Organizations** in the search bar.

Ask Gemini

portal.azure.com/#view/Microsoft_Azure_Resources/AzureDevOps.ReactView

Microsoft Azure

Search resources, services, and docs (G+)

megananth.s.2024.aiml...
DEFAULT DIRECTORY (MEGANA...

Home

Azure DevOps

Azure DevOps

Plan smarter, collaborate better, and ship faster with a set of modern dev services.

View my organizations

Microsoft

Sign out



megananth.s.2024....
megananth.s.2024.aiml@raja...
[My Microsoft account](#)
[Switch directory](#)



Sign in with a different account

https://portal.azure.com/Account/Switch?ru=https%3A%2F%2Fportal.azure.com%2F%23view%2FMicrosoft_Azure_Resources%2FAzureDevOps.ReactView

4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.

The screenshot displays the Azure DevOps Organizations page. The browser's address bar shows the URL `aex.dev.azure.com/me?mkt=en-GB`. The page header includes several tabs: 'Azure DevOps - Microsoft', 'My Information', 'Summary - Overview', 'ChatGPT Images 2.0 | AI', and 'Ask Gemini'. The user profile on the left is for 'Megananth S' with email `megananth.s.2024.aiml@rajalakshmi.edu.in`. The main section is 'Azure DevOps Organizations' with a 'Create new organization' button. It lists three organizations: 'dev.azure.com/megananth2024aiml (Owner)', 'dev.azure.com/abrinrithish (Member)', and 'dev.azure.com/mukeshdharman (Member)'. The first organization has three projects: 'crime rate prediction', 'crime rate', and 'crime-rate -prediction'. An 'Actions' column shows 'Open in Visual Studio' for the first organization. At the bottom, there is a 'Visual Studio Dev Essentials' banner.

RESULT:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2 DATE:	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
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AIM:-

To set up an Azure DevOps project for efficient collaboration and agile work management.

1. Create An Azure Account

2. Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:

Name: Choose a name for the project (e.g., **LMS**).

Description: Optionally, add a description to provide more context about the project.

Visibility: Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

Create new project



Project name *

Batch Data Analysis and Visualization 1

Description

Visibility

 **Private**

Only people you give access to will be able to view this. Want to create a public project? [Try GitHub](#)

^ Advanced

Version control ?

Git



Work item process ?

Agile



Cancel

Create

- Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

Browser tabs: Azure DevOps - Microsoft, My Information, Summary - Overview, ChatGPT Images 2.0 | AI, Ask Gen

Address bar: aex.dev.azure.com/me?mkt=en-GB

Microsoft account: **Megananth S** (rajalakshmi.edu.in)

Visual Studio Dev Essentials: Get everything you need to build and deploy

Azure DevOps Organizations

[Create new organization](#)

▼ [dev.azure.com/megananth2024aiml](#) (Owner)

Projects

- crime rate prediction
- crime rate
- crime-rate -prediction
- [New project](#)

Actions

[Open in Visual Studio](#)

▼ [dev.azure.com/abrinrithish](#) (Member)

▼ [dev.azure.com/mukeshdharman](#) (Member)

4. Project dashboard

dev.azure.com/kamaleshwarank2024aiml/Batch%20Data%20Analysis%20and%20Visualization

Azure DevOps kamaleshwarank2024aiml / Batch Data Analysis and Vis... / Overview / Summary

Batch Data Analysis and Visualization

Private [Invite](#)

About this project

Help others to get on board!
Describe your project and make it easier for other people to understand it.

[+ Add Project Description](#)

Project stats [Period: Last 7 days](#)

Boards

0 Work items created	0 Work items completed
----------------------	------------------------

Repos

0 Pull requests opened	0 Commits by 0 authors
------------------------	------------------------

Members 5

KW PH [3 icons]

Project settings

5. To manage user stories:

- a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.
- b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

The screenshot shows the Azure DevOps interface for the 'crime-rate -prediction Team'. The left-hand navigation menu is visible, with 'Boards' selected. The main area displays the 'Backlog' view, showing a list of work items. The work items are organized into two main sections: '1' and '2'. Section '1' includes an Epic 'Crime Data Management' with a Feature 'Crime Data Collection' and several User Stories. Section '2' includes an Epic 'Crime Prediction System' with a Feature 'Prediction Model Training' and several User Stories. The right-hand sidebar shows the 'Planning' section, which includes a 'crime-rate -prediction Team backlog' and three iterations: 'Iteration 1' (Planned Effort: 0, 18 hours), 'Iteration 2' (No work scheduled yet), and 'Iteration 3' (No work scheduled yet). A '+ New Sprint' button is visible at the bottom right of the sidebar.

Order	Work Item Type	Title	State	Effort	Business Area	Value Area
1	Epic	Crime Data Management	New		Business	
	Feature	Crime Data Collection	New		Business	
	User Story	As an admin, I want to upload crime datasets so th...	New		Business	
	User Story	As an admin, I want to validate uploaded data so th...	New		Business	
	User Story	As an admin, I want to update crime records so tha...	New		Business	
	Feature	Data Preprocessing	New		Business	
	User Story	As a system, I want to clean missing values so that ...	New		Business	
	User Story	As a system, I want to normalize crime data so that...	New		Business	
	User Story	As a system, I want to categorize crime types so th...	New		Business	
2	Epic	Crime Prediction System	New		Business	
	Feature	Prediction Model Training	New		Business	
	User Story	As a developer, I want to train the model using hist...	New		Business	
	User Story	As a developer, I want to evaluate model accuracy ...	New		Business	
	User Story	As a developer, I want to save the trained model so ...	New		Business	
	Feature	Crime Rate Prediction	New		Business	
	User Story	As a user, I want to enter location details so that I c...	New		Business	

dev.azure.com/kamaleshwaran2024aiml/Batch%20Data%20Analysis%20and%20Visualization%20Team/_backlogs/backlog/Batch%20Data%20Analysis%20and%20Visualization%20Team

Batch Data Analysis and Visualization Team Team

Backlog Analytics

Order	Work Item Type	Title	State	Story...	Value Area	Iteration Path	Tags
1	User Story	As a data analyst, I want to upload batch data files (CSV/Exc...	New		Business	Batch Data Analysis and Visualiza...	
2	User Story	As a system user, I want validation rules for batch uploads, ...	New		Business	Batch Data Analysis and Visualiza...	
3	User Story	As a data engineer, I want scheduled batch ingestion, So th...	New		Business	Batch Data Analysis and Visualiza...	
4	User Story	As a data engineer, I want to clean and transform batch dat...	New		Business	Batch Data Analysis and Visualiza...	
5	User Story	As a system, I want to store processed batch data securely, ...	New		Business	Batch Data Analysis and Visualiza...	
6	User Story	As a system user, I want validation rules for batch uploads, ...	New		Business	Batch Data Analysis and Visualiza...	
7	User Story	As a data analyst, I want to perform statistical analysis on b...	New		Business	Batch Data Analysis and Visualiza...	
8	User Story	As a data analyst, I want to apply filters and aggregations, S...	New		Business	Batch Data Analysis and Visualiza...	
9	User Story	As a system, I want to store processed batch data securely, ...	New		Business	Batch Data Analysis and Visualiza...	
10	User Story	As a business user, I want summarized insights from batch a...	New		Business	Batch Data Analysis and Visualiza...	
11	User Story	As a system, I want to automatically generate insight report...	New		Business	Batch Data Analysis and Visualiza...	
12	User Story	As a data engineer, I want to clean and transform batch dat...	New		Business	Batch Data Analysis and Visualiza...	
13	User Story	As a business user, I want interactive dashboards, So that I c...	New		Business	Batch Data Analysis and Visualiza...	
14	User Story	As a business user, I want different chart types (bar, pie, line...	New		Business	Batch Data Analysis and Visualiza...	
15	User Story	As a data analyst, I want to perform statistical analysis on b...	New		Business	Batch Data Analysis and Visualiza...	
16	User Story	As a user, I want to export reports in PDF/Excel format, So t...	New		Business	Batch Data Analysis and Visualiza...	
17	User Story	As an admin, I want role-based access to reports, So that da...	New		Business	Batch Data Analysis and Visualiza...	
18	User Story	As a data analyst, I want to apply filters and aggregations, S...	New		Business	Batch Data Analysis and Visualiza...	

Project settings

Microsoft Sign out

Kamalesh waran
kamaleshwaran.k.2024.aiml@r...
[My Microsoft account](#)
[Switch directory](#)

Sign in with a different account

Result:

setup.

Successfully created an Azure DevOps project with user story management and agile workflow

EXP NO:
3 DATE:

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

AIM:-

To learn about how to create epics, user story, features, backlogs for your assigned project.

The screenshot displays the Azure DevOps interface for a project named 'crime-rate -prediction'. The main view is the 'Backlog' for the 'crime-rate -prediction Team'. The backlog is organized into two main sections, 1 and 2. Section 1 contains an Epic 'Crime Data Management' with a Feature 'Crime Data Collection' and several User Stories. Section 2 contains an Epic 'Crime Prediction System' with a Feature 'Prediction Model Training' and several User Stories. The right sidebar shows the 'Planning' section with iterations 1, 2, and 3.

Order	Work Item Type	Title	State	Effort	Business Area
1	Epic	Crime Data Management	New		Business
	Feature	Crime Data Collection	New		Business
	User Story	As an admin, I want to upload crime datasets so that...	New		Business
	User Story	As an admin, I want to validate uploaded data so that...	New		Business
	User Story	As an admin, I want to update crime records so that...	New		Business
	Feature	Data Preprocessing	New		Business
	User Story	As a system, I want to clean missing values so that...	New		Business
	User Story	As a system, I want to normalize crime data so that...	New		Business
	User Story	As a system, I want to categorize crime types so that...	New		Business
2	Epic	Crime Prediction System	New		Business
	Feature	Prediction Model Training	New		Business
	User Story	As a developer, I want to train the model using hist...	New		Business
	User Story	As a developer, I want to evaluate model accuracy ...	New		Business
	User Story	As a developer, I want to save the trained model so ...	New		Business
	Feature	Crime Rate Prediction	New		Business
	User Story	As a user, I want to enter location details so that I c...	New		Business

Create Epic, Features, User Stories, Task

crime-rate -prediction

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

crime-rate -prediction Team

New Work Item

View as Board

Column Options

Backlog

Analytics

	Order	Work Item Ty...	Title	State	Effort	Busi...	Value Area
		User Story	As a developer, I want to train the model using hist...	New			Business
		User Story	As a developer, I want to evaluate model accuracy ...	New			Business
		User Story	As a developer, I want to save the trained model so ...	New			Business
		Feature	Crime Rate Prediction	New			Business
		User Story	As a user, I want to enter location details so that I c...	New			Business
		User Story	As a user, I want to view predicted crime rates so t...	New			Business
		User Story	As a user, I want to compare past and predicted cri...	New			Business
3		Epic	Visualization and Alerts	New			Business
		Feature	Crime Data Visualization	New			Business
		User Story	As a user, I want to see crime statistics in charts so...	New			Business
		User Story	As a user, I want to filter crime data by location so t...	New			Business
		User Story	As a user, I want to view crime trends over time so ...	New			Business
	+	Feature	Alert and Reporting System	New			Business
	+	User Story	As a user, I want to receive alerts for high crime are...	New			Business
		User Story	As an admin, I want to generate crime prediction re...	New			Business
		User Story	As a user, I want to download crime reports so that ...	New			Business

Epics

Planning

crime-rate -prediction Team

Iteration 1

Planned Effort: 0

18

Iteration 2

No work scheduled yet

Iteration 3

No work scheduled yet

+ New

1.

RESULT:-

Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4 DATE:	SPRINT PLANNING
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AIM:-

To assign user story to specific sprint for the Student Management System Project.

Sprint Planning:-

The screenshot displays the Azure DevOps Wiki interface for a project named 'crime-rate--prediction'. The main content area is titled 'Functional Requirements' and lists seven system requirements for a crime rate prediction system. The requirements are:

- The system shall allow users to upload crime datasets for analysis.
- The system shall preprocess and clean crime data before prediction.
- The system shall train a machine learning model using historical crime records.
- The system shall predict future crime rates based on user input and past data.
- The system shall display crime statistics and prediction results in graphical format.
- The system shall allow users to search and filter crime data by location and crime type.
- The system shall generate downloadable crime prediction reports for users and admins.
- The system shall send alerts or notifications for areas with high predicted crime rates.

The page also shows a search bar, a 'Project settings' link, and a '+ New page' button. The left sidebar contains a navigation menu with options like Overview, Summary, Dashboards, Wiki, Boards, Repos, Pipelines, Test Plans, and Artifacts. The 'Wiki' section is currently selected.

Sprint 1

Sprint 2

crime-rate-prediction

Overview

Summary

Dashboards

Wiki

Boards

Repos

Pipelines

Test Plans

Artifacts

crime-rate--prediction.wiki

Enter page title

Functional Requirements

Non-Functional Requirements

Project settings

New page

Instant Search: You can instantly search wiki pages by activating the search box.

Try it!

dev.azure.com/megananth2024aiml/crime-rate%20-prediction/_wiki/wikis/crime-rate--prediction.wiki/3/Non-Fun...

Search

School

Non-Functional Requirements

Unfollow 1 Edit

Megananth S Just now

- The system should provide prediction results within a few seconds.
- The system should support secure storage and handling of crime data.
- The system should be available 24/7 with minimal downtime.
- The system should handle large datasets efficiently without performance issues.
- The system should provide an easy-to-use and responsive user interface.
- The system should maintain prediction accuracy with reliable machine learning models.
- The system should allow scalability for future data and feature expansion.
- The system should ensure compatibility across desktop and mobile devices.

0 visits in last 30 days

Comments

MS

Add a comment...

RESULT:-

The Sprints are created for the Student Management System.

EXP NO:
5 DATE:

POKER ESTIMATION

AIM:-

Create Poker Estimation for the user stories -Student management System.

Poker Estimation:-

The screenshot shows the Azure DevOps Planning Poker interface for a team game titled "Megananth's Team game". The current item being estimated is "As an admin, I want role-based access to reports. So that data security is maintained." The interface displays a "Reveal" button and a row of seven cards with the values 0, 1, 2, 3, 5, 8, and 13. Below the cards is a "Reference items" section. The "Game scope" table shows the current item as a "New" state with no estimates. On the right, a detailed view of "USER STORY 94" is shown, indicating it is "Unassigned" and has a description that is empty.

Work item	State	Estimates
As an admin, I want role-based access to reports. So that data security is maintained. USER STORY 94	New	-

RESULT:-

The Estimation/Story Points is created for the project using Poker Estimation.

EXP NO:
6 DATE:

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

AIM:-

To Design a Class Diagram and Sequence Diagram for the given Project.

The screenshot shows the Azure DevOps interface for a project named 'crime-rate-prediction'. The left sidebar contains a 'Project Settings' menu with options like Overview, Teams, Permissions, Notifications, Service hooks, Dashboards, Boards, Project configuration, Team configuration, GitHub connections, Pipelines, Agent pools, Parallel jobs, Settings, Test management, Service connections, and XAML build services. The main content area is titled 'Boards' and indicates the project is using the Agile process. It provides instructions on creating and managing iterations for sprint planning. Below this, there is a table of iterations.

Iterations	Start Date	End Date
crime-rate -prediction	...	
Sprint 1	02/02/2026	02/11/2026
sprint 2	03/06/2026	05/06/2026
sprint 3	05/08/2026	16/08/2026

6A. CLASS DIAGRAM:-

6B. SEQUENCE DIAGRAM:-

The screenshot displays the Azure DevOps Planning Poker interface. The browser tabs include "Azure DevOps - Microsoft Az...", "Crime Rate Prediction Epics", and "Planning Poker - Boards". The URL bar shows the path to the specific game. The left sidebar lists various project management tools, with "Planning Poker" currently selected. The main area is titled "Poker game, 11 May 2026" and shows the current item being discussed: "As a developer, I want to evaluate model accuracy so that the best algorithm can be selected". Below this, a row of cards represents the votes, with the number 8 highlighted as "Your vote". A "Reveal" button is present. To the right, a detailed view of "USER STORY 18" is shown, including its description and a "Description is empty" note. At the bottom, a "Game scope" table lists the items and their estimated values.

Game scope

Work item	State	Estimates
As a system, I want to clean missing values so that prediction accuracy improves USER STORY 12	New	5
As a system, I want to categorize crime types so that analysis becomes easier. USER STORY 14	New	3
As a developer, I want to evaluate model accuracy so that the best algorithm can be ... USER STORY 18	New	-

Result:

The Class Diagram and Sequence Diagram is designed Successfully for given project.

EXP NO: 8 DATE:	TESTING – TEST PLANS AND TEST CASES
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AIM:-

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

**Test Planning and Test Case
Test Case Design Procedure****1. Understand Core Features of the Application**

- User Signup & Login
- Student Enrollment in Courses
- Assignment Submission & Evaluation
- Viewing Grades and Attendance
- Admin Role Management and Report Generation

2. Define User Interactions

- Each test case simulates a real user behavior (e.g., logging in, submitting an assignment, viewing results).

3. Design Happy Path Test Cases

- Focused on validating that all core functionalities work correctly under normal conditions.
- **Example:** Student registers and logs in, enrolls in a course, submits assignment, and views grades.

4. Design Error Path Test Cases

- Simulate invalid inputs, system issues, or failed actions to ensure proper error handling.
- **Example:** Login with wrong credentials, submission without attachment, unauthorized access to admin panel.

5. Break Down Steps and Expected Results

- Each test case includes a clear sequence of actions and expected results.
- Ensures both manual testers and automation tools can follow the process easily.

6. Use Clear Naming and IDs

- Test cases are uniquely identified (e.g., TC01 – Student Login Success, TC12 – Invalid Assignment Submission).
- Facilitates easy mapping to features and tracking in Azure DevOps.

7. Separate Test Suites

- Grouped by functionality such as:

- Login and Registration
- Course Enrollment

- Assignment Submission
- Report Generation
- Admin Functions
- Improves organization and enables focused execution in Azure DevOps.

8. Prioritize and Review

- High-priority assigned to critical workflows like login, course access, and grading.
- Reviewed for completeness, accuracy, and alignment with user stories and feature definitions.

The screenshot displays the Azure DevOps web interface. The left sidebar shows the navigation menu with 'Test Plans' selected. The main area shows the 'Crime-Rate-Tests' test suite (ID: 34) in the 'Execute' tab. The test suite contains three test points, with the second one, 'Login with invalid credentials', highlighted in blue and marked as 'Failed'.

Title	Outcome	Order	Test Case Id
☐ login with invalid password	Passed	1	35
☒ Login with invalid credentials	Failed	2	36
☐ Login with invalid username	Passed	3	37

1. Test suite

2. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Student Management System– Test Plans

USER STORIES:-

- As a student, I want to sign up and log in securely so I can access academic services. (ID: 101)
- As a student, I need to enroll in courses to participate in classes. (ID: 102)
- As a student, I should be able to submit assignments online for evaluation. (ID: 103)
- As a student, I want to view grades and attendance in one place. (ID: 104)
- As an admin, I want to manage users and generate performance reports. (ID: 105)

Test Suites:-

Test suit : TS01 – User Login & Registration (ID: 201)

1. TC01 – Successful Student Sign-Up

- **Action:**
 - Go to the Sign-Up page
 - Enter valid name, email, and password
 - Click "Sign Up"
- **Expected Result:**
 - Student account is created and redirected to dashboard
- **Type:** Happy Path

2. TC02 – Login with Valid Credentials

- **Action:**
 - Go to Login page
 - Enter valid student email & password
 - Click "Login"
- **Expected Result:**
 - Logged in and taken to student dashboard
- **Type:** Happy Path

3. TC03 – Sign-Up with Existing Email

- **Action:**
 - Go to Sign-Up
 - Enter already registered email
 - Click "Sign Up"
- **Expected Result:**
 - Error message: "Email already registered"
- **Type:** Error Path

4. TC04 – Login with Incorrect Password

- **Action:**
 - Enter wrong password
 - Click "Login"
- **Expected Result:**
 - Error: "Invalid username or password"
- **Type:** Error Path

Test Suite: TS02 – Course Enrollment (ID: 202)

1. TC05 – Enroll in Available Course

- **Action:**
 - Login → Navigate to Course List
 - Select a course → Click "Enroll"
- **Expected Result:**
 - Enrollment confirmation shown
- **Type:** Happy Path

2. TC06 – Try Enrolling in a Full Course

- **Action:**
 - Select a course with no seats available
 - Click "Enroll"
- **Expected Result:**
 - Error: "Enrollment failed – course full"
- **Type:** Error Path

Test Suite: TS03 – Assignment Submission (ID: 203)

1. TC07 – Submit Assignment Successfully

- **Action:**
 - Login → Go to Assignments
 - Upload file and click "Submit"
- **Expected Result:**
 - Submission confirmation message shown
- **Type:** Happy Path

2. TC08 – Submit Without Uploading File

- **Action:**
 - Open assignment submission
 - Click "Submit" without file
- **Expected Result:**
 - Error: "Please upload a file before submitting"
- **Type:** Error Path

Test Suite: TS04 – View Grades & Attendance (ID: 204)

1. TC09 – View Grades Dashboard

- **Action:**
 - Login → Go to "My Grades"
- **Expected Result:**
 - All grades are displayed by subject
- **Type:** Happy Path

2. TC10 – Fail to Fetch Grades (Offline Scenario)

- **Action:**
 - Disconnect network → Go to "My Grades"
- **Expected Result:**
 - Error: "Unable to fetch data – check connection"
- **Type:** Error Path

Test Suite: TS05 – Admin Management (ID: 205)

1. TC11 – Assign Role to New User

- **Action:**
 - Login as Admin → Navigate to User Management
 - Select user and assign role
- **Expected Result:**
 - Confirmation: "Role assigned successfully"
- **Type:** Happy Path

2. TC12 – Generate Report with No Data

- **Action:**
 - Navigate to Reports → Choose empty semester
 - Click "Generate"
- **Expected Result:**
 - Message: "No data available to generate report"
- **Type:** Error Path

3. Test Cases

Batch Data Analysis an...

Overview

Boards

Repos

Pipelines

Test Plans

Test plans

Progress report

Runs

Exploratory sessions

Artifacts

Batch Data Ingestion (5)

Batch Data Ingestion (ID: 59)

Execute Chart

TEST CASE 61

TC0_01 Login with valid credentials

Kamalesh waran

0 Comments Add Tag

Save

Refresh

Help

Design

Reason

Area

Batch Data Analysis and Visualization

Batch Data Analysis and Visualization/Sprint 1

Updated by Kamalesh waran Apr 6

Steps

Summary

Associated Automation

Recent test results

Deployment

Development

Batch Data Analysis an...

Overview

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Batch Data Ingestion (5)

Batch Data Ingestion (ID: 59)

Execute Chart

Search Title, Test case id and Tags columns

Outcome

Current Tester

Configuration

Assigned To

State

Test Points (5 items)

Title	Outcome	Order	Test Case Id	State	Configuration	Current Tester
☒ TC0_01 Login with valid credentials	Passed	1	61	Design	Windows 10	Kamalesh waran
☐ TC_02 Login with invalid password	Passed	2	62	Design	Windows 10	Kamalesh waran
☐ TC_03 Login with invalid username	Passed	3	63	Design	Windows 10	Kamalesh waran
☐ TC_04 Login with empty username	Passed	4	64	Design	Windows 10	Kamalesh waran
☐ TC_05 Password Field Masked Check	Passed	5	65	Design	Windows 10	Kamalesh waran

Azure DevOps

kamaleshwarank2024aiml / Batch Data Analysis and Vis... / Test Plans / Batch Data Ingestion

Batch Data Analysis an... +

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Batch Data Ingestio...

Feb 2 - Feb 16

Pass

100% run, 100% passed. [View report](#)

Test Suites

Filter suites by name

Batch Data Ingestion (5)

Batch Data Ingestion (ID: 59)

Execute Chart

TEST CASE 61

61 TC0_01 Login with valid credentials

Kamalesh waran 0 Comments Add Tag

Save

Help

State Design Area Batch Data Analysis and Visualization

Reason New Iteration Batch Data Analysis and Visualization(Sprint 1)

Updated by Kamalesh waran: Apr 6

Steps Summary Associated Automation

Steps

Steps Action Expected result

1. Open login page User is redirected to dashboard

2. 2. Enter valid username move to password

3. 3. Enter valid password verify the password

4. Click Login+ open website

Click or type here to add a step

Recent test results

Passed by with Windows 10 Completed 8:36 AM, 4/19/2026

Failed by with Windows 10 Completed 8:36 AM, 4/19/2026

Passed by with Windows 10 Completed 8:36 AM, 4/19/2026

Passed by with Windows 10 Completed 8:36 AM, 4/19/2026

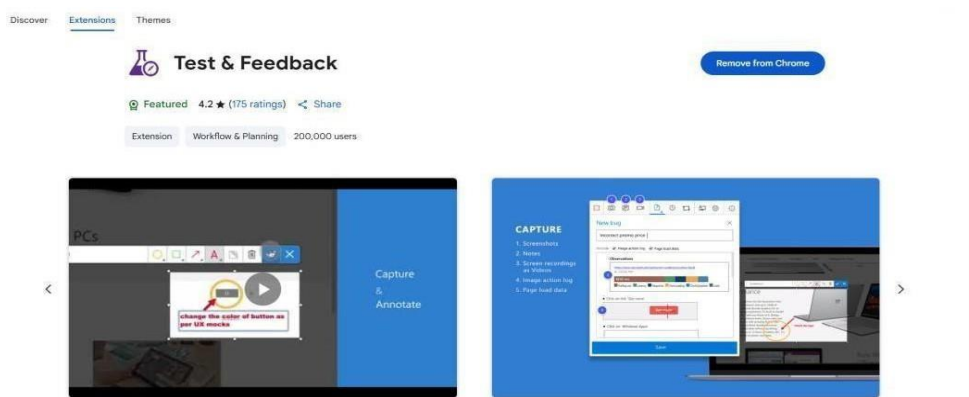
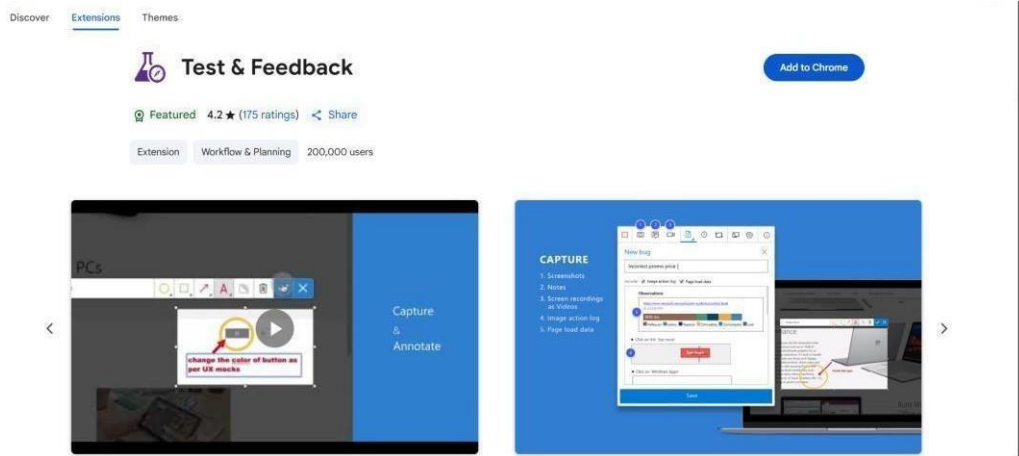
[View all test result history](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

4. Installation of test



5. Running the test cases

Batch Data Analysis an... +

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Batch Data Ingestion (ID: 59)

Feb 2 - Feb 16 **Past**

100% run, 100% passed. [View report](#)

Test Suites

Filter suites by name

Batch Data Ingestion (5)

Batch Data Ingestion (ID: 59)

Execute Chart

Test Points (5 items)

Title	Outcome	Order	Test Case Id	State	Configuration	Current Tester
☒ TC_01 Login with valid credentials	Passed	1	61	Design	Windows 10	Kamalesh waran
☐ TC_02 Login with invalid password	Passed	2	62	Design	Windows 10	Kamalesh waran
☐ TC_03 Login with invalid username	Passed	3	63	Design	Windows 10	Kamalesh waran
☐ TC_04 Login with empty username	Run for web application	64	Design	Windows 10	Kamalesh waran	
☐ TC_05 Password Field Masked Check	Run for desktop application	65	Design	Windows 10	Kamalesh waran	

Save and close | Create bug

44*: Verify whether you can add new users to the application.

1. open the website in the browser

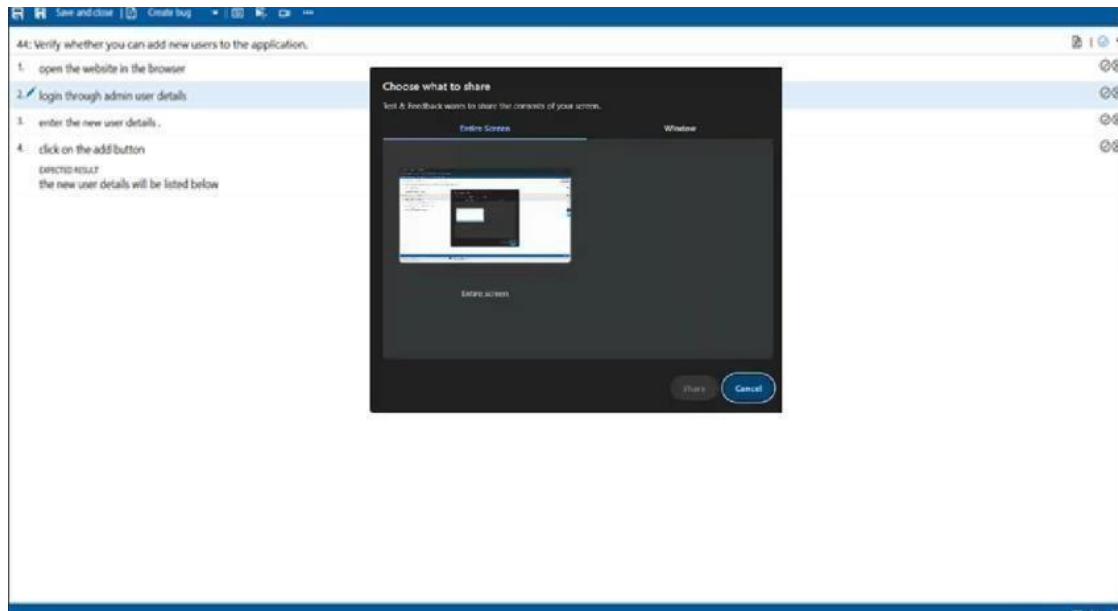
2. login through admin user details

3. enter the new user details .

4. click on the add button

EXPECTED RESULT
the new user details will be listed below

6. Recording the test case



7. Creating the bug

Batch Data Analysis an... +

Recently updated Back to Work Items 5 of 34

BUG 39

39 Invalid Credentials

No one selected 0 Comments Add Tag

Save Follow

Updated by Prasanth H: Monday

Details

System Info

Browser - Name	Google Chrome 146
Browser - Language	en-GB
Browser - Height	1032
Browser - Width	1920
Browser - User agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Operating system - Name	Windows NT 10.0; Win64; x64
Operating system - Architecture	x86_64
Operating system - Processor model	12th Gen Intel(R) Core(TM) i9-13900
Operating system - Number of processors	32
Memory - Available	22793551872
Memory - Capacity	34033346000
Display - Pixels per inch (X axis)	96
Display - Pixels per inch (Y axis)	96
Provider - Platform model name	i9

Resolved Reason

Story Points

Priority

2

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Test Results

Azure DevOps

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Batch Data Analysis and Vis...

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Search

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Batch Data Ingesti... ▾ ⭐

Feb 2 - Feb 16 Pass

100% run, 100% passed. [View report](#)

Test Suites   

Filter suites by name

Batch Data Ingestion (5)

Batch Data Ingestion (ID: 59)

Help

Execute Chart    

Search Title, Test case id and Tags columns Outcome ▾ Current Tester ▾ Configuration ▾ Assigned To ▾ State ▾ X

Test Points (5 items)  Run for web application ▾

 Title	Outcome	Order	Test Case Id	State	Configuration	Current Tester
☒ TC0_01 Login with valid credentials	Passed	1	61	Design	Windows 10	Kamalesh waran
☐ TC_02 Login with invalid password	Passed	2	62	Design	Windows 10	Kamalesh waran
☐ TC_03 Login with invalid username	Passed	3	63	Design	Windows 10	Kamalesh waran
☐ TC_04 Login with empty username	Passed	4	64	Design	Windows 10	Kamalesh waran
☐ TC_05 Password Field Masked Check	Passed	5	65	Design	Windows 10	Kamalesh waran

8. Test report summary

dev.azure.com/kamaleshwarank2024aiml/Batch%20Data%20Analysis%20and%20Visualization/_testPlans/execute?planId=58&suiteId=59

Batch Data Ingestion (ID: 59)

Test Suites

Batch Data Ingestion (5)

Test Points (5 items)

- ☒ TC0_01 Login with valid credentials
- ☐ TC_02 Login with invalid password
- ☐ TC_03 Login with invalid username
- ☐ TC_04 Login with empty username
- ☐ TC_05 Password Field Masked Check

Test point ID 2 execution history

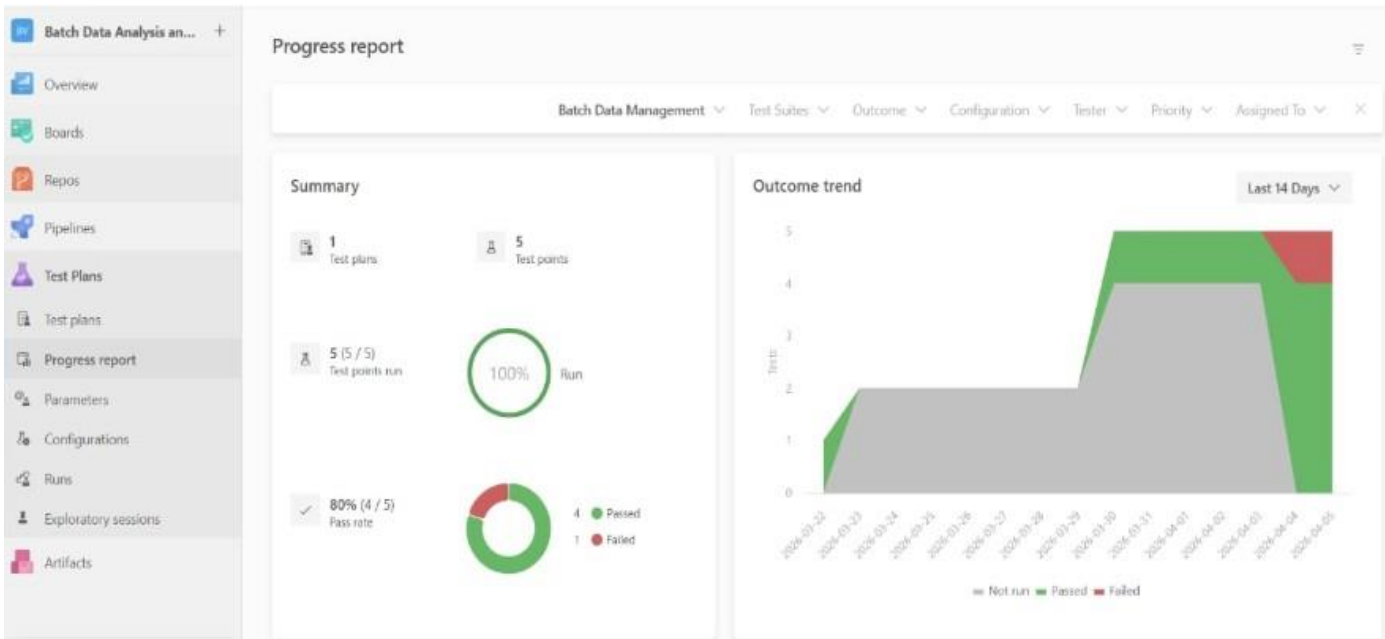
Test Case: 61 TC0_01 Login with valid credentials
Test Suite: 59 Batch Data Ingestion
Current Tester: Kamalesh waran
Configuration: Windows 10

Outcome	Test run	Run by	Time completed
Passed	17	Kamalesh waran	Apr 19
Failed	16	Kamalesh waran	Apr 19
Passed	14	Kamalesh waran	Apr 19
Passed	13	Kamalesh waran	Apr 19
Not Applicable	12	Kamalesh waran	Apr 19
Passed	4	Kamalesh waran	Apr 6

Currently showing a test point execution history. View all history for a full test case execution history. [View all history](#)

1. Assigning bug to the developer and changing state

9. Progress report



10. Changing the test template

 Azure DevOps

kamaleshwarank2024aiml / Settings / Process

Organization Setti...

kamaleshwarank2024aiml

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Filter by process name

Name	Description	Team projects
Basic (default)	This template is flexible for any process and great for teams getting started with Azure DevOps.	1
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	2
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record of decisions.	0

Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9 DATE:	Ci/CD PIPELINES IN AZURE
----------------------------------	---------------------------------

AIM:

To implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline in Azure DevOps for automating the build, testing, and deployment process of the Student Management System, ensuring faster delivery and improved software quality.

PROCEDURE:

Steps to Create and implement pipelines in Azure:

1. Sign in to Azure DevOps and Navigate to Your Project
Log in to dev.azure.com, select your organization, and open the project where your Student Management System code resides.
2. Connect a Code Repository (Azure Repos or GitHub)
Ensure your application code is stored in a Git-based repository such as Azure Repos or GitHub. This will be the source for triggering builds and deployments in your pipeline.
3. Create a New Pipeline
Go to the Pipelines section on the left panel and click “Create Pipeline”.
Choose your source (e.g., Azure Repos Git or GitHub), and then select the repository containing your project code.
4. Choose the Pipeline Configuration
You can select either the YAML-based pipeline (recommended for version control and automation) or the Classic Editor for a GUI-based setup.
If using YAML, Azure DevOps will suggest a template or allow you to define your own.
5. Define Build Stage (CI - Continuous Integration) from YAML file
6. Install dependencies (e.g., npm install, dotnet restore)
7. Build the application (dotnet build, npm run build)
8. Run unit tests (dotnet test, npm test)
9. Publish build artifacts to be used in the release stage

10. Save and Run the Pipeline for the First Time

Save the YAML or build definition and click “Run”.

Azure will fetch the latest code and execute the defined build and test stages.

11. Configure Continuous Deployment (CD)

Navigate to the Releases tab under Pipelines and click “New Release Pipeline”.

Add an Artifact (from the build stage) and create a new Stage (e.g., Development, Production).

12. Configure the CD stage with deployment tasks such as deploying to Azure App Service, running database migrations or scripts, and restarting services using the Azure App Service Deploy task linked to your subscription and app details.

13. Set Triggers and Approvals

Enable continuous deployment trigger so the release pipeline runs automatically after a successful build.

For production environments, configure pre-deployment approvals to ensure manual verification before release.

14. Monitor Pipelines and Manage Logs

View all pipeline runs under the Runs section.

Check logs for build/test/deploy stages to debug any errors.

You can also integrate email alerts or Microsoft Teams notifications for build failures.

15. Review and Maintain Pipelines

Regularly update your pipeline tasks or YAML configurations as your application grows.

Ensure pipeline runs are clean and artifacts are stored securely.

Integrate quality gates and code coverage policies to maintain code quality.

crime-rate -prediction

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TEST CASE 35

35 login with invalid password

Megananth S

0 Comments

Add Tag

Save

Follow

State

Design

Area

crime-rate -prediction

Reason

New

Iteration

crime-rate -prediction\sprint 2

Steps

Recent test results

Deployment

Updated by Megananth S: 22m ago

Steps

Summary

Associated Automation

0

0

1. open login pagin

2. Expected Result

3. Go to the Page

Click or type here to add a step

State...

Action

Expected result

A

Parameter values

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

RESULT:

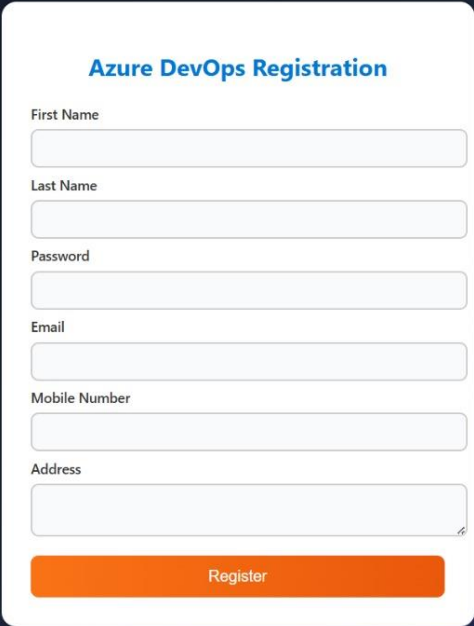
Thus the pipelines for the given project “Student Management System has been executed successfully

EXP NO: 10
DATE:

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Student Management System project.

A registration form for Azure DevOps is displayed on a dark blue background. The form is white with rounded corners and contains the following fields: First Name, Last Name, Password, Email, Mobile Number, and Address. Each field has a light gray input box. Below the fields is an orange 'Register' button. The title 'Azure DevOps Registration' is in blue text at the top of the form.

GitHub Project Structure

Result:

The GitHub repository clearly displays the organized project structure and consistent naming

conventions, making it easy for users and contributors to understand and navigate the codebase.